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PCB Service

- FR-4 PCBs
- Flexible PCBs
- Metal Core PCBs
- High-Frequency PCBs

PCBA Service

- PCB Assembly
- Parts Library

More

- SMT Stencil
- PCB Layout

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- JLC3DP - 3D Printing
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- JLCMC - Mechanical Components

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Capabilities Support

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About Us

JLCPCB delivers one-stop electronic and mechanical manufacturing solutions to empower hardware innovators worldwide.

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PCB Assembly Capabilities

Comprehensive PCB assembly capabilities for SMT, THT, and mixed-technology builds with rigorous inspection for prototypes, NPI, and volume production.

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PCBA Capabilities

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PCB Assembly Technical Capabilities

Features	Economic PCBA	Standard PCBA
Assembly Types	Single sided placement (SMT/Thru-hole)	Single & double sided placement (SMT/Thru-hole)
PCB Layer	2,4,6-layers	1 - 32 layers
Thickness	0.8mm - 1.6mm	No limit
Dimension	Single PCB Size: 10x10mm - 470x500mm PCB Panel Size: 10x10mm - 250x250mm	Single PCB Size: 70x70mm - 460x500mm PCB Panel Size: 70x70mm - 250x250mm
Order Volume	2 - 50 pcs	2 - 80000 pcs
Surface Finish	Limited by specific options (Refer to the options for Economic PCBA in the table below)	No limit
PCB Color	Limited by specific options (Refer to the options for Economic PCBA in the table below)	No limit
Delivery Format	Single PCB, Panel with mouse bites	Single PCB, Panel with mouse bites, Panel with V-cut

Layer Stackup	Standard stack-up only, special stack-up is not supported	All stack-up
Gold Fingers/Castellated Holes/Edge Plating	Not Support	Support
Edge Rails	Not necessary	Necessary
Fiducials	Not necessary	Necessary
Minimum Package	0402	0201
Minimum IC Pin Spacing	0.4mm	0.35mm
Minimum BGA Spacing	0.5mm(center to center)	0.3mm(center to center)
Reflow temperature	255+/-5 °C (not adjustable)	240+/-5 °C
SPI	No	Yes
AOI	Yes	Yes
Visual Inspection	Yes	Yes
X-ray Inspection	Yes (only for certain parts, such as BGA)	Yes (only for certain parts, such as BGA)
Build time	1 - 3 days	≥ 4 days

PCB Specs for Economic PCB Assembly

Layer	Thickness(mm)	Color	Surface Finish	Qty(pcs)
0.8	Green	Lead-Free:HASL Leaded:HASL	2-30	
1	Green/Black	Lead-Free:HASL Leaded:HASL	2-30	
1.2	Green/Black	Lead-Free:HASL Leaded:HASL	2-30	
1.6	Green	Lead-Free:HASL / ENIG Leaded:HASL / ENIG	2-50	

1.6	Black	Lead-Free:HASL Leaded:HASL	2-50
1.6	Blue/Purple	Lead-Free:HASL Leaded:HASL	5-30
1.6	Red/White	Leaded:HASL	5-30
1	Green	Lead-Free:HASL Leaded:HASL	2-30
1.2	Green	Lead-Free:HASL Leaded:HASL	2-50
1.6	Green	Lead-Free:HASL / ENIG Leaded:HASL / ENIG	2-50
1.6	Green	ENIG	2-30

FAQs about JLCPCB PCB Assembly Capabilities

What are the minimum and maximum board sizes supported for PCBA?

Single PCB Assembly

Standard PCBA: 70 × 70 mm to 460 × 500 mm

Economic PCBA: 10 × 10 mm to 470 × 500 mm

Panelized PCB Assembly

Standard PCBA: 70 × 70 mm to 250 × 250 mm

Economic PCBA: 10 × 10 mm to 250 × 250 mm

For panelized PCBs, larger panels are more susceptible to bending and deformation during SMT processing. Therefore, we recommend keeping panel dimensions within 250 × 250 mm whenever possible.

If your panel slightly exceeds the recommended size and the design requires it, production may still be feasible.

Why is the maximum panel size for PCBA much smaller than the maximum single-board size?

The difference is mainly related to manufacturing reliability and process stability during SMT assembly.

In a panelized design, individual PCBs are connected by breakaway tabs, mouse bites, or V-cuts. As panel size increases, the overall panel becomes heavier and less rigid, making it more vulnerable to stress during conveyor transport, reflow soldering, and other assembly processes.

Excessive panel size may result in:

- Panel flexing or warping
- Reduced placement accuracy
- Soldering defects
- Damage to panel connection points
- Increased manual handling requirements

In some cases, if connection points break during production, additional manual soldering or rework may be required, which can affect both production efficiency and product quality.

For optimal assembly yield and reliability, we recommend keeping panel dimensions within the suggested range. However, we understand that certain designs may require larger panels, so our system allows dimensions beyond the recommended size limits. Customers should evaluate the associated manufacturing risks when designing and ordering oversized panels.

Does JLCPCB charge extra tooling for flex PCB assembly, and are there any placement limitations around bend areas?

Yes. For flex PCB assembly, an additional fixture (tooling) fee applies due to the handling requirements of flexible materials during SMT assembly. The fixture cost is \$23.57 per fixture, and the required quantity depends on your production volume.

For a detailed breakdown of PCBA charges, including flex PCB fixture costs, please refer to:

PCB Assembly Price Guide

JLCPCB's assembly process is fully compatible with standard flex PCB materials. For optimal manufacturability and long-term reliability, avoid placing large or heavy components near bending areas and maintain sufficient copper clearance within flexible regions.

Do you support FPC (Flexible PCB) assembly?

Yes. JLCPCB provides assembly services for FPCs. Dedicated fixtures are typically required to secure the boards during assembly. Please refer to the flex PCB manufacturing capabilities.

What is the maximum PCB panel size for assembly?

For optimal SMT process stability and soldering yield, the recommended maximum panel size is **250 × 250 mm**. Assembled PCBs are typically shipped with manufacturing rails (tooling edges) attached by default. If you require depanelization before shipment, you may select the panel removal service when placing your order.

Can you assemble THT components? Do you support mixed assembly (SMD + THT)?

Yes. JLCPCB supports Through-Hole Technology (THT) component assembly and mixed-technology (SMD + THT) PCB assembly within a single PCBA order.

SMD components are typically assembled using reflow soldering, while THT components are assembled using wave soldering.

This service is available under both the Economic and Standard PCBA.

What happens to unused through-holes during THT assembly? Will they be filled with solder?

No. During wave soldering for through-hole (THT) assembly, solder is applied only to the designated through-hole components included in the assembly order.

Unused through-holes are protected from solder ingress during the soldering process, allowing them to remain available for any components customers may wish to install manually after delivery.

Do you support fine-pitch ICs such as BGA and QFN?

Yes. JLCPCB supports advanced fine-pitch component assembly, including BGA, LGA, QFN, QFP, and other high-density IC packages.

Minimum supported lead pitch:

- **Standard PCBA service:** 0.35 mm

- **Economy PCBA service:** 0.40 mm

Can JLCPCB manufacture and assemble 01005-size components?

Yes. JLCPCB supports the assembly of 01005 package components.

For designs containing components with significantly different package sizes, we strongly recommend evaluating the use of a step stencil. A step stencil helps ensure that each component type receives the appropriate solder paste volume, improving solder joint quality and overall assembly yield.

Do you provide X-ray inspection for hidden solder joints?

Yes. For packages with hidden or inaccessible solder joints—such as BGA, QFN, and LGA—JLCPCB performs X-ray inspection as a required quality control step.

If such components are detected in your BOM, X-ray inspection is automatically applied by the system during production. The associated cost is calculated based on your order configuration and included in the final quotation.

For more details:

- X-ray report interpretation guide

- View X-ray inspection fee details

Can you solder wires to a PCB for a power connection, like this? The wire is 22 AWG, 300 V rated, and 150 mm long. Can I create a part number for this wire?

If this is an assembly order, we can support this service. We will review your request, and normally there will be a small additional charge for this manual soldering service. Please select the wire from our parts library, and include its part number in the remark.

Do you offer MCU programming? Can I send a .hex file to program the MCU before it is soldered onto the PCB?

For standard assembly orders, we partially support this programming service. The customer can select the advanced option and provide the information listed below. Then our engineering team will evaluate whether it is supported.

1. The manufacturer part number of the chip
2. The interface definition reserved on the board for programming
3. An image of the board
4. The programming file in HEX or BIN format

Can JLCPCB assemble stacked components, such as a DIP IC inserted into a DIP socket, or a microcontroller mounted on a pin header/socket assembly?

Please note that we only provide assembly service and do not ship parts separately. Installing an IC part A on a DIP socket part B is acceptable.

To do this, please include both part A and part B in the BOM and CPL.

The coordinate of part B in the CPL should be normal. Then please add one more coordinate for part A, which should be more than 0.2 mm away from part A. In this way, both part A and part B can be selected on the parts selection page when you place the assembly order.

Then please submit the order with a PCBA remark: need to put the B part on the A part.

This process is normally handled manually, so there will be some additional manual operation costs.

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